



数据库第二次作业

```
employee (ID, person_name, street, city)
works (ID, company_name, salary)
company (company_name, city)
manages (ID, manager_id)
```

图 3-19 雇员数据库

3.9 考虑图3-19 的雇员数据库，其中的主码被加了下划线。 请为下面的每个查询写出 SQL表达式：

- 找出为 "First Bank Corporation" 工作的每位雇员的 ID、姓名及所居住城市。
- 找出为 "First Bank Corporation" 工作且工资超过 10 000 美元的每位雇员的 ID、姓名及所居住城市
- 找出没为 "First Bank Corporation" 工作的每位雇员的 ID。
- 找出工资高于 "Small Bank Corporation" 的所有雇员的每位雇员的 ID。
- 假设一家公司可以位于好几个城市。找出位于 "Small Bank Corporation" 所有所在城市的每家公司的名称。
- 找出雇员最多的公司名称（如果雇员最多的不止一家、则全部列出这些公司）。
- 找出平均工资高于 "First Bank Corporation" 平均工资的每家公司的名称

a.

```
select employee.ID, person_name, city
from employee, works
```

```
where employee.ID = works.ID
      and works.company_name = "First Bank Corporation";
```

b.

```
select employee.ID, person_name, city
from employee, works
where employee.ID = works.ID
      and works.company_name = "First Bank Corporation"
      and works.salary > 10000;
```

c.

```
select ID
from works
where ID not in (
      select ID
      from works
      where company_name = "First Bank Corporation"
);
```

d.

```
select ID
from works
where salary > all (
      select salary
      from works
      where company_name = "Small Bank Corporation"
);
```

e.

```
select distinct company_name
from company as c1
where not exists(
      (select city
```

```
from company
where company_name = "Small Bank Corporation")
except
(select city
from company as c2
where c1.company_name = c2.company_name));
```

f.

```
select company_name
from works
group by company_name
having count(ID) >= all (
    select count(ID)
    from works
    group by company_name
);
```

g.

```
select company_name
from works
group by company_name
having avg(salary) > (
    select avg(salary)
    from works
    where company_name = "First Bank Corporation"
);
```

3.16 考虑图3-19 中的雇员数据库。其中主码被加了下划线。 请给出下面每个查询的SQL表达式。

- 找出每位这样的雇员的 ID 和姓名：该雇员所居住的城市与其工作的公司所在城市一样。
- 找出所居住的城市和街道与其经理相同的每位雇员的ID和姓名。
- 找出工资高于其所在公司所有雇员平均工资的每位雇员的ID和姓名。
- 找出工资总和最小的公司。

a.

```
select employee.ID, employee.person_name
from employee, works, company
where employee.ID = works.ID
      and works.company_name = company.company_name
      and employee.city = company.city;
```

b.

```
select e.ID, person_name
from employee as e, manages, employee as m
where e.ID = manages.ID
      and manages.manager_id = m.ID
      and (e.city, e.street) = (m.city, m.street);
```

c.

```
select e.ID, person_name
from employee as e, works as w
where e.ID = w.ID
      and salary > (
        select avg(salary)
        from works
        where works.company_name = w.company_name
      );
```

d.

```
with total_salary(company_name, salary_sum) as (
  select company_name, sum(salary)
  from works
  group by company_name)

select company_name
from total_salary
where salary_sum = (
```

```
select min(salary_sum)
from total_salary
);
```

3.17 考虑图 3-19 中的雇员数据库。请给出下面每个查询的SQL表达式。

a. 为 "First Bank Corporation" 的所有雇员增长 10% 的工资。

```
update works
set salary = salary * 1.1
where company_name = "First Bank Corporation";
```

b. 为 "First Bank Corporation" 的所有经理增长 10% 的工资。

```
update works
set salary = salary * 1.1
where company_name = "First Bank Corporation"
and ID in (
    select manager_id
    from manages
);
```

c. 删除 "Small Bank Corporation" 的雇员在 works 关系中的所有元组。

```
delete from works
where company_name = "Small Bank Corporation";
```